EXP. NO: 2
23/7/26

STUDY OF NETWORK CABLES

AIM:

Study of different types of Network cables

a) Understand different types of network cable

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial cable
4. Fibre optic cable

Cable type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application use
UTP	Category 3	10 Mbps	Advantages: * cheaper in cost * Easy to install as they have a smaller overall diameter Disadvantages: * More prone to (EMI) Electromagnetic Interference & noise	10 Base-T Ethernet
	Category 5	upto 100 Mbps		Fast Ethernet, Gigabit Ethernet
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet
STP	Category 6, 6a	10 Gbps	Advantages: * Shielded * Faster than UTP * Less susceptible to noise and interference Disadvantages: * Expensive * Greater installation effort	Gigabit Ethernet, 10G Ethernet (55 m) widely used in data centres
SSTP	Category 7	10 Gbps		Gigabit Ethernet 10G Ethernet (100 m)

Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages: * High bandwith * Immune to interference * Low loss bandwith * Versatile Disadvantages: * Expensive * Greater installation effort	Speed of signal is 500m Television network High speed internet connections
fibre optics cable	Single mode multi mode	100 Gbps	Advantages: * High speed * High bandwith * High security * Long distance Disadvantages: * Expensive * Requires skilled installers	Maximum distance of fibre optics cable is around 100 meters

b) Make your own ethernet cross-over cable / straight cable

Tools and parts needed:

* Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
* A crimping tool. This is an all-in-one networking tools shaped to push down the pins in the plug and strip and cut the shielding off the cables.
* Two RJ45 plugs.
* optional two plug shields.

Step 1: To start construction of the cable, begin by threading shields onto the cable.

Step 2: Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3: After, you will need to untangle the wires; there should be four "twisted pairs". Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4: Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create and even level. The difficult aspect is placing these into the RJ45 plug without messing up the other. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you, as shown

Step 5: Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means that you stripped off too much shielding, simply snip the cables back a little more.

Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

It should be shaped correctly, but pushing too hard can crack the fragile plastic plug.

Step 7: Lastly, repeat for the other end using diagram B (to make a crossover cable) / using diagram A (to make a straight through cable)

To test it, plug it in and attempt to connect two devices directly.

RESULT:

Thus the different types of network cables are studied successfully.